IMPACT OF SUPER RESOLUTION ON THE DETECTION OF PLASTIC BOTTLES IN NATURE

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ABSTRACT

The aim of this project was to enhance detection of plastic wastes in low-res images by using super-resolution methods. First, we simulated low-quality real-world images by downgrading our dataset's HR images. Next, we enhanced degraded images using Real-ESRGAN before object detection. We trained YOLOv11 models independently with HR, LR, and SR images. The performance exhibited enhanced visual quality (SSIM improved from 0.3605 to 0.4436) and detection quality compared to LR (mAP50 from 0.518 to 0.572). Although SR did not match HR performance entirely, it had definite gains, indicating that super-resolution can be applied effectively in detection in low-quality images.

1 INTRODUCTION

Plastic pollution is one of today's most critical environmental problems. Out of all the different kinds of trash, plastic bottles are everywhere and pretty harmful because they last a long time and people use them a lot. Finding plastic bottles in nature is an important step if we want to make cleanup processes automatic and help with tracking pollution.

However in real life, detection systems often use low-quality pictures, such as the ones taken by drones or cameras that are far away. These images can be blurry or noisy, and that makes it hard to see plastic bottles clearly.

The main idea of this project is to see if super-resolution (SR) techniques can make these low-quality images better, and if that actually helps improve how well plastic bottles are detected. We look into using SR before detection to see if it works better than just using the original blurry images.

One of the biggest problems is that low-res images do not have enough detail for the detectors to work properly. Even advanced models such as YOLOv11 have difficulty in spotting small or blurry objects. Training with high-res images usually gives better results, but those kinds of images are not always available.

Super-resolution might help fix this by trying to add back the missing details from the blurry images. That could make detection more accurate and close the gap between low and high-quality images.

Because of how important this issue is for the environment, and because current systems do not work that well with low-res images, we wanted to test if combining SR and object detection is useful. We focus on ESRGAN to see if it can boost image quality enough to help YOLOv11 detect plastic bottles more accurately.

In the end, our project gives some insight into how computer vision could be used to detect plastic waste on a large scale in real-world situations.

2 EXPERIMENTAL SETTINGS

Our first experiments began with a dataset that had plastic bottle images captured from very far distances, mostly using static cameras monitoring the same location over time. As a result, many of the images were extremely similar as they are taken at slightly different times around the same riverbank or landscape, showing minimal changes in bottle positions or lighting. In most of these images, the plastic bottles appeared only as a few small pixels, making them not suitable for both training an effective detection model and for super-resolution enhancement. The invariance and clarity in this dataset significantly limited its usability for our objectives.

[`https://github.com/m0-n/Plastic-Bottles-Dataset/tree/master`]

To get rid of these issues, we switched to a second dataset. It that had much clearer and more diverse images. This dataset had a wide range of scenes, lighting conditions, and bottle appearances, making it more closer to real-world as it had more variability. The plastic bottles were more easier to spot and better suited for both object detection and super-resolution training. As a result, we ended up using this second dataset for the rest of our experiments.

`https://universe.roboflow.com/ukm-wcn/plastic-bottles-in-nature/`

2.1 Dataset Preparation

The dataset we used was originally designed for segmentation tasks, it had annotations in polygon format. Moreover, it contained a number of unlabeled images and empty annotation files, which were not usable for training. To ensure that the dataset is prepared for object detection, we filtered out all images without labels and converted the segmentation-style polygon annotations into bounding box formats that are compatible with YOLO. This process consisted of calculating the bounding boxes from the polygon coordinates and rewriting the annotations accordingly. As a result, we obtained a clean, detection-ready dataset that aligns with the YOLOv11 training pipeline.

3 Method

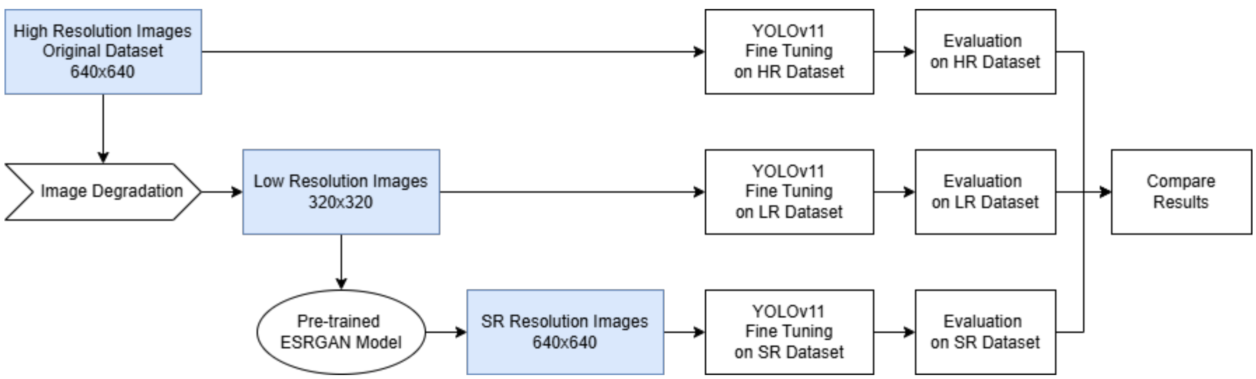


Figure 1: Methodology Pipeline

3.1 Image Degradation

To simulate real-world conditions where images are captured from drones or distant surveillance cameras, we artificially degraded our high-resolution (HR) dataset to create a corresponding low-resolution (LR) version. The original dataset images were 640×640 pixels in size. Each image was downscaled by a factor of 2, resulting in 320×320 resolution images.

The degradation process consisted of three main steps:

1. Downscaling: Bicubic interpolation is used to resize each image to half its original resolution. This is a crucial step for representing the limited detail captured by low-quality or distantly positioned imaging devices.
2. Blurring: A Gaussian blur was applied using a 5×5 kernel and a sigma value of 1.0 to simulate the optical imperfections of real-world lenses.
3. Noise Injection: Gaussian noise with standard deviation of 5 was added to represent the sensor noise that is usually present in real-world images, especially under poor lighting or low-quality hardware conditions.

This preprocessing pipeline allowed us to create a low-resolution dataset that reflects realistic input conditions for super-resolution models. During this process, only the images were altered. The corresponding YOLO-format labels, being normalized coordinates, were directly copied without modification.

As a result, we obtained two datasets:

1. HR Dataset: The original, clean high-resolution images.
2. LR Dataset: The degraded images simulating real-world low-quality inputs.

These datasets served as the basis for training and evaluating both the super-resolution and detection models.

3.2 Super-Resolution with ESRGAN

We used a super-resolution (SR) method called ESRGAN (Enhanced Super-Resolution Generative Adversarial Networks) to improve the quality of low resolution images and potentially increase the detection performance. At first, we tried training a custom ESRGAN model using our own dataset, where the low-res (LR) images were the inputs and the original high-res (HR) images were used as ground truth. However, training from scratch took a long time, and the results while still being okay, were not better than what we got from a pretrained ESRGAN model.

Thus, we ended up using the pretrained Real-ESRGAN x2 model, which was both faster and gave better quality results. This model upscaled the LR images back to 640×640 resolution, which matched the original HR size. If any output image did not exactly match this size, we resized it manually. We also kept the YOLO-format annotations from the LR dataset since the image scaling was consistent, so the labels were still accurate.

This process produced a third dataset, SR-enhanced, containing super-resolved images. In total, we worked with three datasets throughout our experiments:

1. HR Dataset: Original high-resolution images (640×640)
2. LR Dataset: Artificially degraded images (320×320)
3. SR Dataset: Super-resolved images (restored to 640×640)

These datasets enabled us to assess the effect of image resolution and enhancement on object detection performance using consistent and comparable setups.

(a) Low Resolution (b) Custom ESRGAN Super Resolution (c) Real-ESRGANx2 Super Resolution

Figure 2: Comparison between low resolution image and different ESRGAN generated images

3.3 Object Detection

We used YOLOv11, a modern and efficient object detection model, for the detection part of our project. Our goal was to see how well plastic bottles could be detected under three different image conditions: high-resolution (HR), low-resolution (LR), and super-resolved (SR). To make the comparison fair, we trained and tested a separate YOLOv11 model on each dataset using their own train, validation, and test splits.

We carefully matched the input resolution of the model to the image type to ensure that each model worked within the resolution it was designed for, that way, scaling of the images did not interfere with the fairness of the evaluation:

- HR and SR images were trained and evaluated using an input size of 640×640, maintaining their resolution.
- LR images were trained and evaluated using an input size of 320×320, consistent with their degraded format.

Each model was fine-tuned for 50 epochs using the Adam optimizer, a learning rate of 0.001, and a batch size of 16. A confidence threshold of 0.25 and an IoU threshold of 0.7 is used for both training and evaluation. The training used pre-trained weights (yolo11n.pt) as a base to help the model learn faster and improve generalization.

After the training process had ended, models were evaluated using the validation and test sets with performance metrics such as mean Average Precision (mAP), precision, and recall. The prediction results were stored for future analysis.

By having this kind of experimental design, we were allowed us to assess the impact of image resolution and super-resolution enhancement on detection accuracy, robustness, and inference efficiency.

4 Experimental Results

4.1 Training Setup and Hyperparameters

YOLOv11n is used as the object detection model for all the experiments that was done.

Each model was trained for 50 epochs, with a batch size of 16, using the Adam optimizer and an learning rate of 0.001. The selection of the hyperparameters followed the best practices in the literature. They are known for offering stable convergence and strong generalization, especially in our case since we are training a smaller model (YOLOv11n) on a medium-sized dataset.

- For HR and SR datasets, we used an input size of 640×640, which matches their native resolution.
- For the LR dataset, we used a reduced input size of 320×320, to match with its downsampled size.

4.2 Training Curves

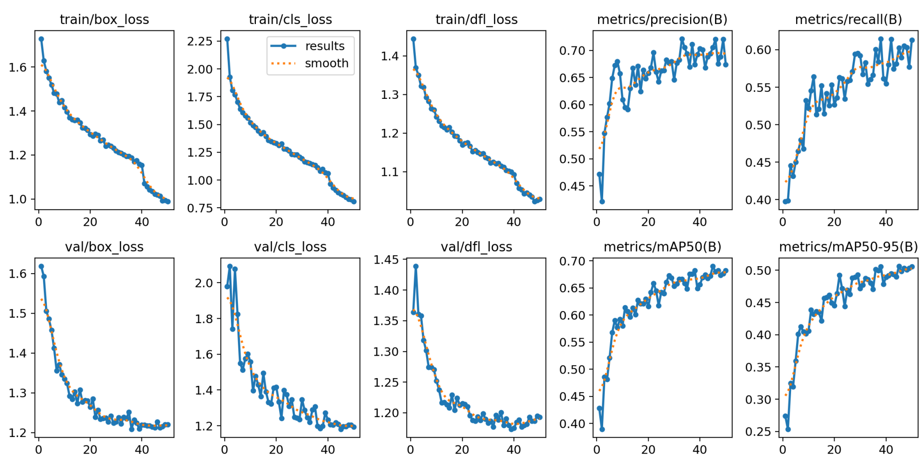


Figure 3: High Resolution Curves

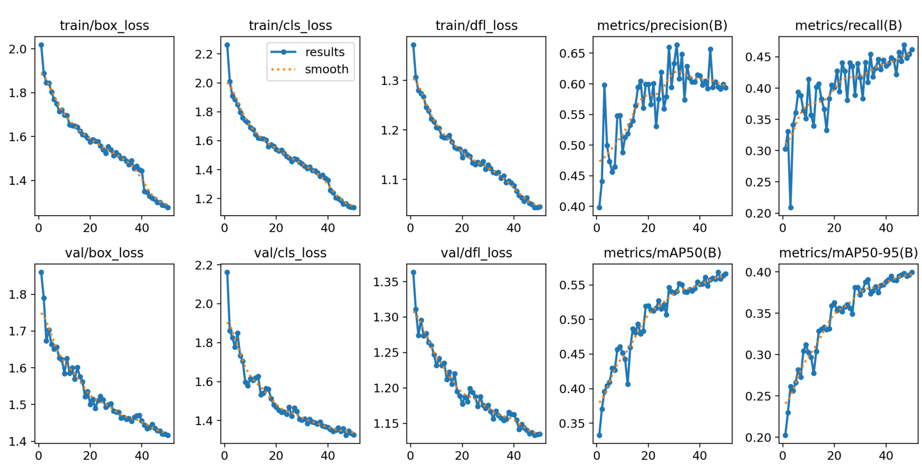


Figure 4: Low Resolution Curves

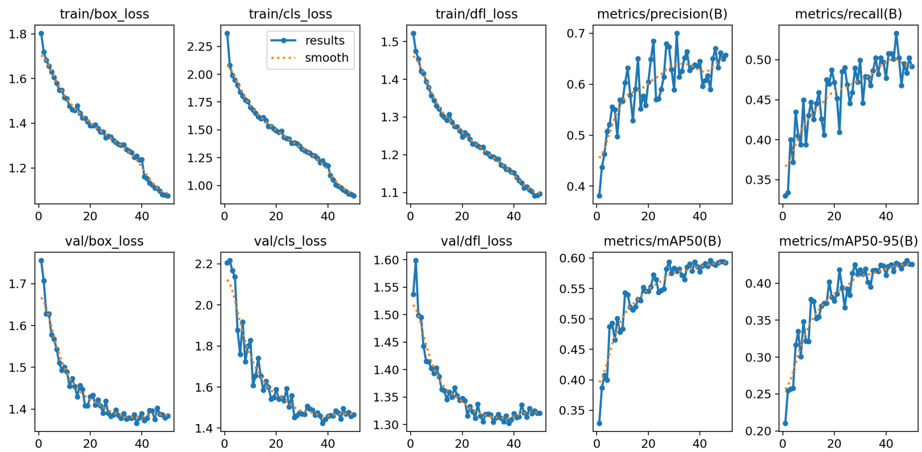


Figure 5: Super Resolution Curves

4.3 Quantitative Evaluation

We examined visual similarity and detection metrics across the datasets to determine how super-resolution affected image quality and object detection performance.

4.3.1 Image Quality Metric

We first measured the Structural Similarity Index Measure (SSIM) to quantify the perceptual quality improvements that super-resolution (SR) introduced. SSIM compares luminance, contrast, and structure between two images, which makes it well suited to evaluate image restoration quality.

We computed the average SSIM:

- Between High-Resolution (HR) and Low-Resolution (LR) images.
- Between High-Resolution (HR) and Super-Resolved (SR) images.

	Average SSIM
HR - LR	0.3605
HR - SR	0.4436

Table 1: SSIM Values of HR - LR/SR

As shown, SR images achieved a higher similarity to the original HR images, which shows super-resolution recovered visual details lost during degradation.

4.3.2 Evaluation Metrics

	Precision	Recall	mAP50	mAP50-95
High Resolution	0.748	0.546	0.671	0.499
Low Resolution	0.561	0.418	0.518	0.373
Super Resolution	0.744	0.411	0.572	0.418

Table 2: Performance Between Datasets

These results support the following key observations:

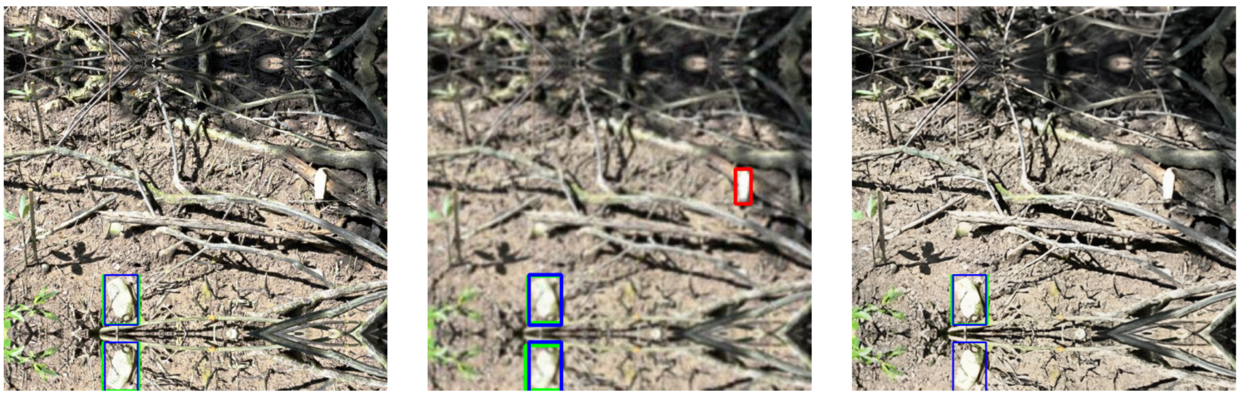
- The Low Resolution dataset had the poorest performance at all the metrics, as would be expected with low image resolution.

- All measures in Super Resolution data had a substantial gain over LR, especially mAP@0.50 (+5.4%) and mAP@0.50:0.95 (+4.5%), suggesting that object detectability was enhanced by SR.
- The accuracy of the SR model (0.744) was very close to that of the HR model (0.748), suggesting super-resolved images were visually competent enough for the model to produce accurate predictions.
- Even while recall in SR was a little behind HR, overall gain relative to LR is impressive, reinforcing the fact that super-resolution improves detection.

4.4 Qualitative Results

In Figures 6, 7, 8, 9, there can be seen images from our three datasets and bounding boxes predicted from their corresponding YOLOv11n model. The blue color represent ground truth bounding boxes, green color represent true predictions' bounding boxes and red color represent false predictions' bounding boxes.

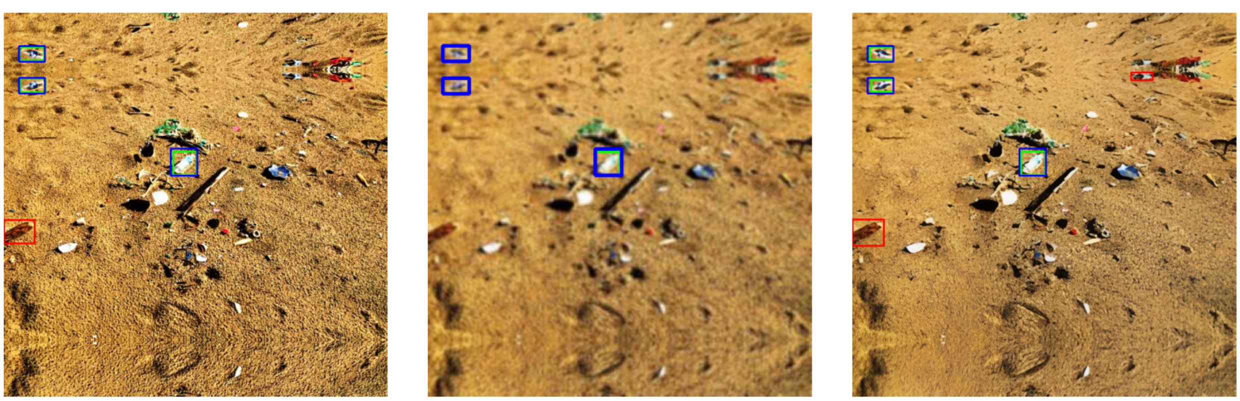
In Figure 6, it can be seen that the SR image missed a ground truth annotation which HR and LR catched. This can be an evidence image small recall difference between LR and SR in Table 2.



(a) High Resolution (b) Low Resolution (c) Super Resolution

Figure 6: Super Resolution misses a ground truth label

Figure 7 demonstrates that super-resolution enhances object detection accuracy on low-resolution inputs, increasing the number of correctly detected objects.



(a) High Resolution (b) Low Resolution (c) Super Resolution

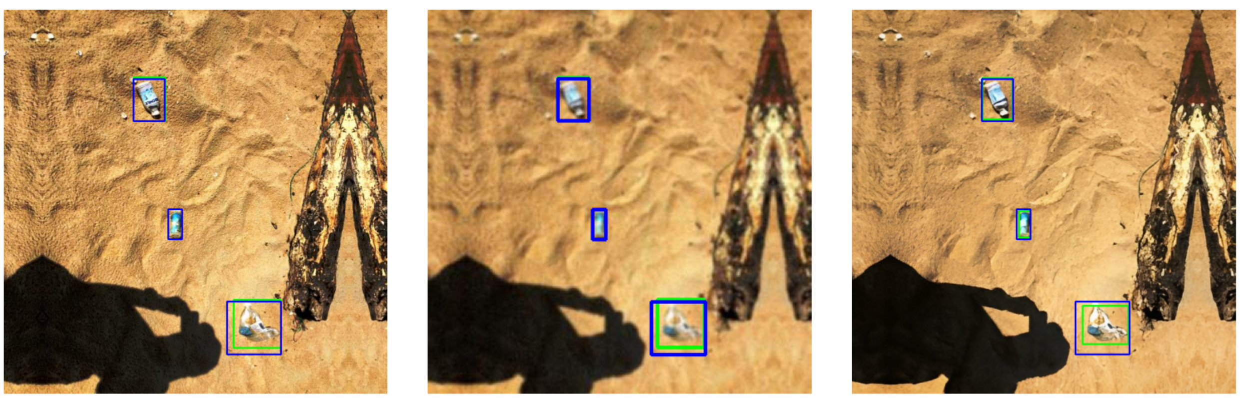
Figure 7: Super resolution improves the true positive rate of low resolution image

Figure 8 shows that the SR-enhanced image reduces the number of incorrect detections present in the LR version, bringing it closer to HR performance.

(a) High Resolution (b) Low Resolution (c) Super Resolution

Figure 8: Super resolution improves the false positive rate of low resolution image

Figure 9 shows that SR sometimes helps recover missed detections even in HR images, decreasing the false negative rate.



(a) High Resolution (b) Low Resolution (c) Super Resolution

Figure 9: Super resolution improves the false negative rate of high resolution image

4.5 Runtime and Efficiency

4.5.1 Runtime on Super Resolution

We initially attempted to utilize the dataset to train our own ESRGAN model to generate super resolution images. However, the process took a very long time, taking approximately 8 hours to complete 200 training epochs.

Meanwhile, the Real-ESRGAN pretrained model gave even better performance while significantly reducing processing time to approximately 20 minutes. We solely utilized the pretrained model for the rest of our project due to its quality and efficiency.

	Running Time
Our Own ESRGAN	8 hours
Pretrained ESRGAN	20 minutes

Table 3: Running Times of Training and Using a Pretrained ESRGAN Model

4.5.2 Runtime on Model Training

The training time for the object detection model (YOLOv11n) varied based on the resolution of the images:

	Running Time
High Resolution	30 minutes
Low Resolution	20 minutes
Super Resolution	29 minutes

Table 4: Running Times of YOLOv11n Training on Different Datasets

As expected, low-resolution (320×320) images trained faster due to smaller input sizes. Training on super-resolved images (640×640) took almost the same time as high-resolution training.

5 Discussions / Conclusions

As a whole, the project was a success. Our objective was to see if super-resolution (SR) would help improve object detection performance from low-quality plastic waste images, specifically from far away or from small objects. The outcome was that using ESRGAN greatly improved both perceived image quality (SSIM) and detection performance (mAP) over the low resolution baseline.

5.1 Strengths

Among the best assets in our pipeline was our utilization of Real-ESRGAN, a model that processed degraded images quickly and efficiently. The pretrained model outperformed our own trained model and also granted us a quick means to produce super-resolved images at great quality with minimal overhead. We also made a balanced comparison by training YOLOv11 independently for HR, LR, and SR data with suitably scaled input sizes, providing us with a neat setup for evaluation.

5.2 Limitations

Nevertheless, the SR score in the SR dataset was still low, a little above the LR baselines. This indicates that although SR makes it easier to see objects, it fails to recover perfectly enough fine details to enable robust detection in every instance. Our SR model was also constrained by not having domain-specific training, something that would likely have facilitated better adaptation to plastic waste texture and designs.

5.3 Improvement Opportunities

Future enhancements might consist of:

- Increasing relevance in SR outputs by training the ESRGAN model directly using images of plastic waste.
- Comparing with other SR techniques such as SwinIR or EDSR.
- Employing a larger or more various dataset to generalize more effectively across settings.
- Using data augmentation or training multi-scale to enable detection models to better cope with variability.

5.4 Challenges and Solutions

A key hurdle was discovering a useful data-set. The initial data-set we attempted, mentioned in section 2 experimental setup, had major drawbacks (very small objects, repetitive views) and thus was not useful. We resolved this by switching to a better data-set quality and developing preprocessing scripts for label cleaning and annotation conversion.

An additional major challenge was extensive training periods for ESRGAN; switching to a pretrained model resulted in a lot of time savings without compromising quality.

Super-resolution has been shown to improve detection in low-quality images.

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